

MCS-061 Mathematical Foundations-I

Solved question bank · June 2026 paper, fully worked, with practice drills

Sets

Q1. If a set A is defined as $A = \{x : x \text{ is a prime number and } 1 < x < 10\}$, calculate the total number of elements in its power set $P(A)$. From this total, how many are considered as proper subset? [5 marks · Jun 2026 Q1(a)]

Step 1 — list the set explicitly. The primes strictly between 1 and 10 are 2, 3, 5, 7. $A = \{2, 3, 5, 7\}$, so the cardinality is $n = |A| = 4$.

Step 2 — formula for the power set. For a finite set with n elements, the power set $P(A)$ is the set of all subsets of A and $|P(A)| = 2^n$.

Step 3 — substitute. $|P(A)| = 2^4 = 16$.

Step 4 — proper subsets. A proper subset of A is any subset $S \subseteq A$ with $S \neq A$. Of the 16 subsets, exactly one (the set A itself) is not proper, so number of proper subsets $= 2^n - 1 = 16 - 1 = 15$.

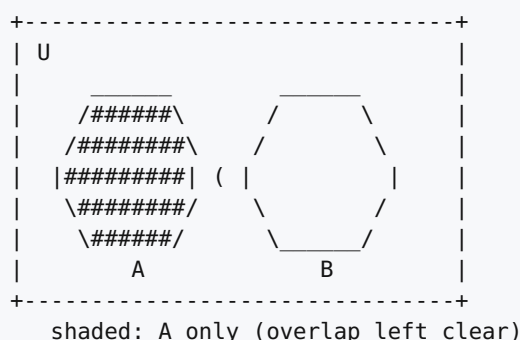
Answer: $|P(A)| = 16$, and 15 of these are proper subsets.

Justification of the counts: each of the 4 elements is independently either in a subset or out of it, giving $2 \times 2 \times 2 \times 2 = 16$ selections, hence 2^n . The subtraction of exactly 1 removes only the subset $A = \{2, 3, 5, 7\}$; the empty set $\emptyset$ is still counted as a proper subset, because $\emptyset \subseteq A$ and $\emptyset \neq A$.

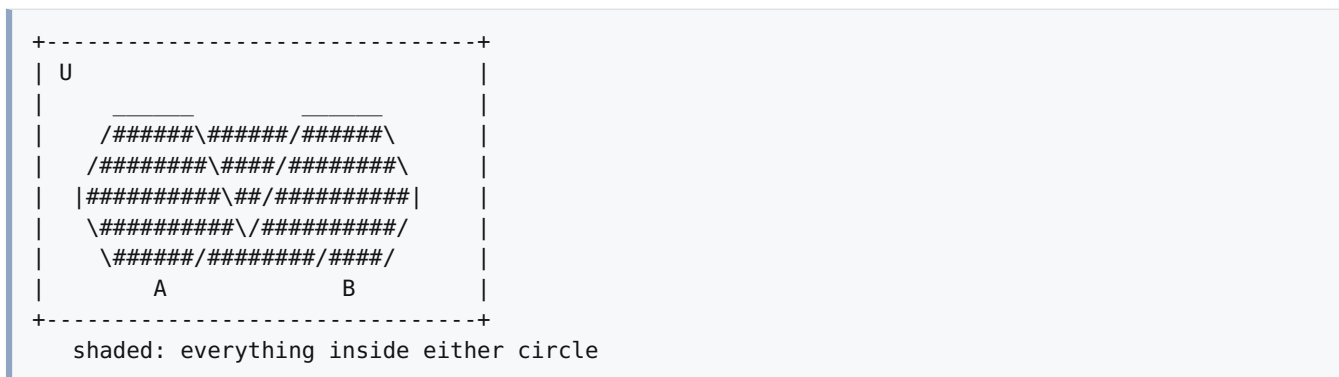
Q2. Draw Venn diagrams for the following operations between set A and set B : (i) $A - B$ (ii) $A \cup B$ (iii) $A \cap B$ [6 marks · Jun 2026 Q3(b)]

In every diagram the rectangle is the universal set U and the two overlapping circles are A (left) and B (right). Only the required region is shaded (shown here by ###).

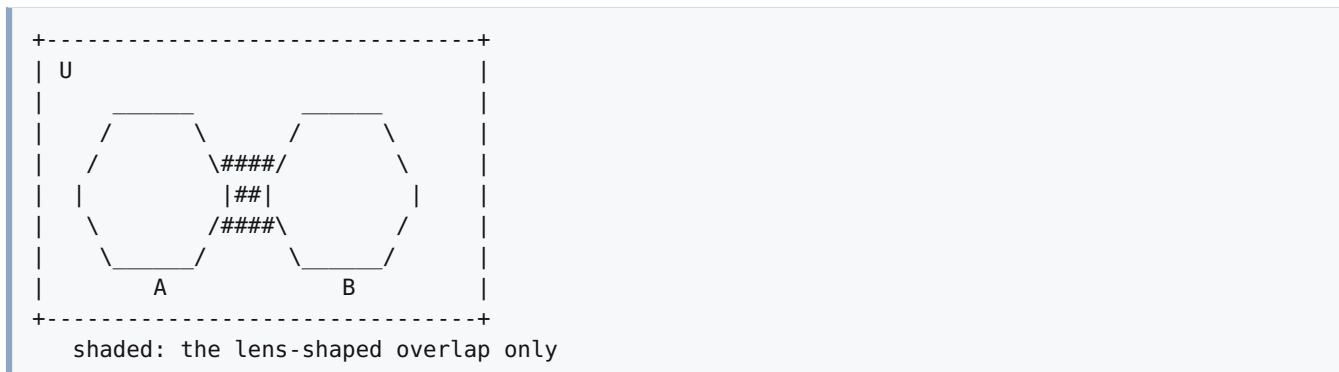
(i) $A - B = \{x : x \in A \text{ and } x \notin B\}$ — the part of A outside B .



(ii) $A \cup B = \{x : x \in A \text{ or } x \in B \text{ or both}\}$ — both circles entirely.



(iii) $A \cap B = \{x : x \in A \text{ and } x \in B\}$ — the overlap only.



Answer: (i) the region of A excluding the overlap; (ii) the whole of both circles; (iii) the overlap region only.

Naming each region, as the marks require: in (i) the shaded part contains elements of A that are not in B; in (ii) the shaded part contains every element belonging to at least one of A, B; in (iii) the shaded part contains exactly the elements common to A and B. The unshaded area outside both circles in every diagram is $(A \cup B)'$.

Q3. $B = \{x : x \text{ is an even number and } 0 < x < 12\}$. Find $|P(B)|$, the number of proper subsets, and the number of non-empty proper subsets. [practice]

Step 1 — list B. $B = \{2, 4, 6, 8, 10\}$, so $n = 5$.

Step 2 — apply the formulas. $|P(B)| = 2^5 = 32$. Proper subsets = $2^n - 1 = 32 - 1 = 31$ (excluding B itself). Non-empty proper subsets = $2^n - 2 = 32 - 2 = 30$ (excluding B itself and $\emptyset$).

Answer: $|P(B)| = 32$, proper subsets = 31, non-empty proper subsets = 30.

Reason for each subtraction: the first -1 removes B, which is a subset but not a proper one; the second -1 removes $\emptyset$, which is a proper subset but is empty.

Q4. $A = \{x : x \text{ is a factor of } 12\}$. List A, give $|P(A)|$, and state how many subsets of A have exactly 2 elements. [practice]

Step 1 — list the factors of 12. $A = \{1, 2, 3, 4, 6, 12\}$, so $n = 6$.

Step 2 — power set size. $|P(A)| = 2^6 = 64$.

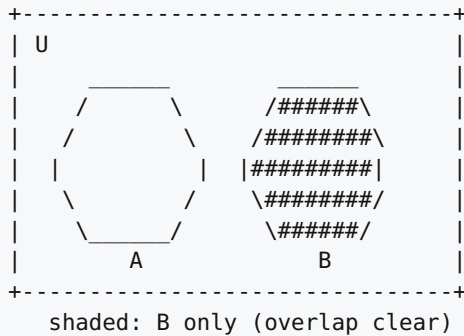
Step 3 — subsets of size exactly 2. A subset of size 2 is an unordered selection of 2 objects from 6, so the count is the combination $C(6, 2) = 6!/(2! \times 4!) = (6 \times 5)/2 = 15$.

Answer: $A = \{1, 2, 3, 4, 6, 12\}$, $|P(A)| = 64$, and 15 subsets have exactly 2 elements.

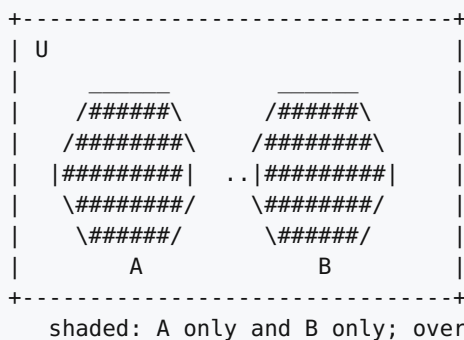
Reason: subsets are unordered, so $\{1, 2\}$ and $\{2, 1\}$ are the same subset — hence a combination $C(6, 2)$ and not a permutation.

Q5. Draw Venn diagrams for (i) $B - A$ (ii) $A \triangle B$ (symmetric difference) (iii) $(A \cup B)'$, shading each. [practice]

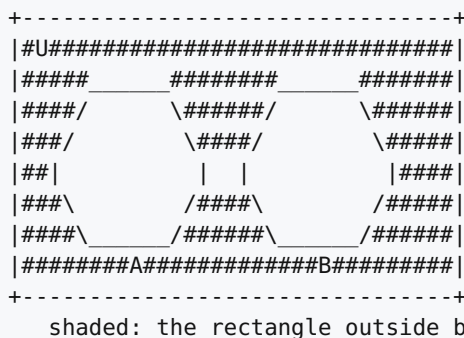
(i) $B - A = \{x : x \in B \text{ and } x \notin A\}$ — the part of B outside A.



(ii) $A \triangle B = (A - B) \cup (B - A)$ — both circles except the overlap.



(iii) $(A \cup B)' = U - (A \cup B)$ — everything outside both circles.



Answer: (i) B excluding the overlap; (ii) both circles with the overlap excluded; (iii) the region of U outside both circles.

Interpretation: $A \triangle B$ holds the elements in exactly one of the two sets, so $A \triangle B = (A \cup B) - (A \cap B)$; $(A \cup B)'$ holds the elements of U in neither set, and by De Morgan's law $(A \cup B)' = A' \cap B'$.

Q6. $A = \{1, 2, 3, 4, 5\}$, $B = \{4, 5, 6, 7\}$. Compute $A - B$, $A \cup B$, $A \cap B$, $A \triangle B$ and verify $|A \cup B| = |A| + |B| - |A \cap B|$. [practice]

$A - B$ = elements of A not in $B = \{1, 2, 3\}$. $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$. $A \cap B = \{4, 5\}$. $A \triangle B = (A - B) \cup (B - A) = \{1, 2, 3\} \cup \{6, 7\} = \{1, 2, 3, 6, 7\}$.

Verification by the inclusion-exclusion principle: $|A| = 5$, $|B| = 4$, $|A \cap B| = 2$. $|A| + |B| - |A \cap B| = 5 + 4 - 2 = 7$. Direct count: $|A \cup B| = |\{1, 2, 3, 4, 5, 6, 7\}| = 7$. $7 = 7$, so the identity holds.

Answer: $A - B = \{1, 2, 3\}$, $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$, $A \cap B = \{4, 5\}$, $A \triangle B = \{1, 2, 3, 6, 7\}$; $|A \cup B| = 7 = 5 + 4 - 2$.

Why the $-|A \cap B|$ is needed: adding $|A|$ and $|B|$ counts every element of the overlap twice, so one copy of each of the 2 common elements must be subtracted to count the union once each. That is exactly the inclusion-exclusion principle.

Relations

Q1. Given a finite set A with 3 elements and a finite set B with 2 elements, calculate the total number of distinct relations that can be formed from A to B . Based on the mathematical definition of a relation, why is the 'empty set' ($\emptyset$) included in this total? [5 marks · Jun 2026 Q1(b)]

Step 1 — definition used. A relation R from A to B is any subset of the Cartesian product $A \times B$, i.e. $R \subseteq A \times B$.

Step 2 — size of the Cartesian product. $|A \times B| = |A| \times |B| = 3 \times 2 = 6$.

Step 3 — count the subsets. The number of relations = number of subsets of $A \times B = 2^{(|A| \times |B|)} = 2^6 = 64$.

Answer: 64 distinct relations can be formed from A to B .

Answer to the rider: the definition places no condition on a relation beyond being a subset of $A \times B$. Each of the 6 ordered pairs is independently either included or excluded, which is what gives 2^6 ; the choice "exclude all six pairs" is one of these 2^6 selections and it produces $R = \emptyset$. Since $\emptyset \subseteq A \times B$ is true for every set, $\emptyset$ satisfies the definition of a relation and is counted — it is called the empty relation (no element of A is related to any element of B). Excluding it would make the count 63 and would contradict the definition.

Q2. Describe the meaning of non-reflexive and anti-reflexive property of any relation, with the help of an example for each. [5 marks · Jun 2026 Q3(c)]

Take the base set $A = \{1, 2, 3\}$ throughout.

Reflexive (the reference point). R on A is reflexive if $(a, a) \in R$ for every $a \in A$. Example: $R = \{(1, 1), (2, 2), (3, 3), (1, 2)\}$ is reflexive.

Non-reflexive. R is non-reflexive if it is simply not reflexive, i.e. the reflexive condition fails for at least one element: there exists some $a \in A$ with $(a, a) \notin R$. Some self-pairs may still be present. Example: $R_1 = \{(1, 1), (2, 2), (1, 3)\}$ on A . Here $(1, 1) \in R_1$ and $(2, 2) \in R_1$, but $(3, 3) \notin R_1$. One self-pair is missing, so R_1 is non-reflexive.

Anti-reflexive (also called irreflexive). R is anti-reflexive if $(a, a) \notin R$ for every $a \in A$ — no element at all is related to itself. Example: $R_2 = \{(1, 2), (2, 3), (1, 3)\}$ on A . None of $(1, 1)$, $(2, 2)$, $(3, 3)$ belongs

to R_2 , so R_2 is anti-reflexive. The relation " $a < b$ " on numbers is the standard anti-reflexive relation, since $a < a$ is never true.

Answer: non-reflexive = the reflexive property fails for at least one element (at least one (a, a) missing); anti-reflexive/irreflexive = $(a, a) \notin R$ for every element, no self-pair at all.

Why the distinction matters: anti-reflexive is a strictly stronger condition than non-reflexive. Every anti-reflexive relation is non-reflexive, but not conversely — R_1 above is non-reflexive yet not anti-reflexive, because it does contain $(1,1)$. So the two terms are not synonyms, and a relation may also be neither reflexive nor anti-reflexive.

Q3. $|A| = 4$, $|B| = 3$. How many distinct relations from A to B ? How many from A to A ? How many of those on A are reflexive? [practice]

Relations from A to B : $|A \times B| = 4 \times 3 = 12$, so count = $2^{12} = 4096$.

Relations from A to A : $|A \times A| = 4 \times 4 = 16$, so count = $2^{16} = 65536$.

Reflexive relations on A : a reflexive relation must contain all 4 diagonal pairs $(1,1)$ -type, so those 4 pairs are forced. The remaining $16 - 4 = 12$ pairs are free to be in or out. Count = $2^{12} = 4096$.

Answer: 4096 relations from A to B , 65536 relations on A , of which 4096 are reflexive.

Reason: the general rule is $2^{(|A| \cdot |B|)}$ because a relation is an arbitrary subset of the product set. For the reflexive count, the n diagonal pairs lose their freedom of choice, leaving $2^{(n^2 - n)}$ with $n = 4$.

Q4. $A = \{1, 2, 3\}$. Write one relation on A that is symmetric but not transitive, and one that is transitive but not symmetric. Justify each in a line. [practice]

Definitions used. Symmetric: $(a, b) \in R \rightarrow (b, a) \in R$. Transitive: $(a, b) \in R$ and $(b, c) \in R \rightarrow (a, c) \in R$.

(i) Symmetric but not transitive: $R = \{(1, 2), (2, 1), (2, 3), (3, 2)\}$. Symmetric: every pair appears with its reverse — $(1,2)$ with $(2,1)$, and $(2,3)$ with $(3,2)$. Not transitive: $(1, 2) \in R$ and $(2, 3) \in R$, but $(1, 3) \notin R$. One failing instance is enough.

(ii) Transitive but not symmetric: $R = \{(1, 2), (2, 3), (1, 3)\}$. Transitive: the only chain is $(1,2)$ with $(2,3)$, and its closure $(1,3)$ is present; there is no other pair $(a,b),(b,c)$ to check. Not symmetric: $(1, 2) \in R$ but $(2, 1) \notin R$.

Answer: symmetric-not-transitive $R = \{(1,2), (2,1), (2,3), (3,2)\}$; transitive-not-symmetric $R = \{(1,2), (2,3), (1,3)\}$.

Why this shows independence: symmetry and transitivity are logically independent properties, so a relation can satisfy either one alone; demonstrating one counterexample pair in each direction is the whole proof.

Q5. Define symmetric, anti-symmetric and asymmetric relations, with one example each on $A = \{1, 2, 3\}$. [practice]

Symmetric. For all a, b : if $(a, b) \in R$ then $(b, a) \in R$. Example: $R = \{(1, 2), (2, 1), (3, 3)\}$ — the reverse of every pair is present.

Anti-symmetric. For all a, b : if $(a, b) \in R$ and $(b, a) \in R$ then $a = b$. Equivalently, two distinct elements are never related both ways; self-pairs are allowed. Example: $R = \{(1, 1), (1, 2), (2, 3)\}$ — no pair of distinct elements occurs in both orders. The relation " $a \leq b$ " is anti-symmetric.

Asymmetric. For all a, b : if $(a, b) \in R$ then $(b, a) \notin R$. This forbids (a, a) as well, since (a, a) would be its own reverse. Example: $R = \{(1, 2), (2, 3), (1, 3)\}$ — no reversed pair, and no self-pair. The relation " $a < b$ " is asymmetric.

Answer: symmetric — reverse of every pair present; anti-symmetric — both orders only when $a = b$; asymmetric — never both orders, and no self-pairs at all.

The relationship among the three: asymmetric = anti-symmetric + irreflexive, so every asymmetric relation is anti-symmetric but not conversely ($\{(1,1)\}$ is anti-symmetric and not asymmetric). A relation that is both symmetric and anti-symmetric can contain only diagonal pairs.

Q6. R on the set of integers defined by aRb iff $a - b$ is divisible by 3. Show R is an equivalence relation and list its equivalence classes. [practice]

Definition used: R is an equivalence relation iff it is reflexive, symmetric and transitive.

Reflexive. For any integer a , $a - a = 0$ and $0 = 3 \times 0$ is divisible by 3, so aRa for all $a \in \mathbb{Z}$. Reflexive. $\checkmark$

Symmetric. Suppose aRb , so $a - b = 3k$ for some integer k . Then $b - a = -3k = 3(-k)$, and $-k$ is an integer, so 3 divides $b - a$ and bRa . Symmetric. $\checkmark$

Transitive. Suppose aRb and bRc , so $a - b = 3k$ and $b - c = 3m$ with $k, m \in \mathbb{Z}$. Adding, $a - c = (a - b) + (b - c) = 3k + 3m = 3(k + m)$, and $k + m$ is an integer, so 3 divides $a - c$ and aRc . Transitive. $\checkmark$

All three hold, hence R is an equivalence relation.

Equivalence classes. aRb means a and b leave the same remainder on division by 3, and the possible remainders are 0, 1, 2. Therefore there are exactly 3 classes: $[0] = \{\dots, -6, -3, 0, 3, 6, \dots\} = \{3n : n \in \mathbb{Z}\}$ $[1] = \{\dots, -5, -2, 1, 4, 7, \dots\} = \{3n + 1 : n \in \mathbb{Z}\}$ $[2] = \{\dots, -4, -1, 2, 5, 8, \dots\} = \{3n + 2 : n \in \mathbb{Z}\}$

Answer: R is an equivalence relation, with the three equivalence classes $[0]$, $[1]$, $[2]$ — the integers of remainder 0, 1 and 2 modulo 3.

Interpretation: the three classes are pairwise disjoint and their union is $\mathbb{Z}$, so they form a partition of $\mathbb{Z}$. This is the general fact that an equivalence relation on a set partitions that set, and conversely.

Functions

Q1. Given the functions $f(x) = x^2$ and $g(x) = x + 2$, calculate the value of composite function $(g \circ f)(5)$ and $(f \circ g)(5)$. [5 marks · Jun 2026 Q1(c)]

Definition used: $(g \circ f)(x) = g(f(x))$ — apply f first, then g . $(f \circ g)(x) = f(g(x))$ — apply g first, then f .

Computation of $(g \circ f)(5)$: $f(5) = 5^2 = 25$ $(g \circ f)(5) = g(25) = 25 + 2 = 27$

Computation of $(f \circ g)(5)$: $g(5) = 5 + 2 = 7$ $(f \circ g)(5) = f(7) = 7^2 = 49$

General forms as a check: $(g \circ f)(x) = g(x^2) = x^2 + 2 \rightarrow$ at $x = 5$: $25 + 2 = 27$ $\checkmark$ $(f \circ g)(x) = f(x + 2) = (x + 2)^2 \rightarrow$ at $x = 5$: $7^2 = 49$ $\checkmark$

Answer: $(g \circ f)(5) = 27$ and $(f \circ g)(5) = 49$.

Conclusion required by the question's framing: $27 \neq 49$, so $(g \circ f)(5) \neq (f \circ g)(5)$. Composition of functions is therefore not commutative in general — the order in which the functions are applied changes the result, because the output of the inner function becomes the input of the outer one.

Q2. Algebraically demonstrate why the equation of a circle $x^2 + y^2 = 4$ does not correspond to a mathematical function. If you were to apply a "Vertical Line Test" to its graph, what would you observe? [5 marks · Jun 2026 Q1(d)]

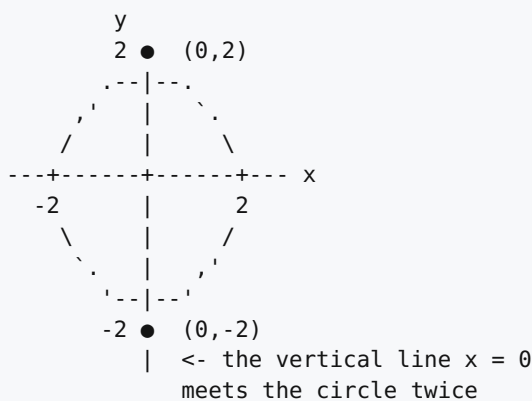
Definition used: a relation is a function of x only if every value of x in the domain is assigned exactly one value of y .

Algebraic demonstration. Solve the equation for y : $x^2 + y^2 = 4 \implies y^2 = 4 - x^2 \implies y = \pm\sqrt{4 - x^2}$

The $\pm$ sign is the whole argument: for one admissible input there are two outputs. Take $x = 0$: $y = \pm\sqrt{4} = +2$ and -2 . The single input $x = 0$ is paired with two different outputs, $(0, 2)$ and $(0, -2)$. Take $x = 1$: $y = \pm\sqrt{3} \approx +1.732$ and -1.732 — again two outputs.

Since at least one input has more than one output, the equation $x^2 + y^2 = 4$ does not define y as a function of x .

Vertical Line Test. A graph represents a function of x iff no vertical line meets it in more than one point. The graph of $x^2 + y^2 = 4$ is a circle of centre $(0, 0)$ and radius 2. Drawing the vertical line $x = 0$ (the y -axis) cuts the circle at $(0, 2)$ and $(0, -2)$ — two points. Indeed every vertical line $x = c$ with $-2 < c < 2$ cuts the circle twice.



Answer: $x^2 + y^2 = 4$ is not a function of x , because solving gives $y = \pm\sqrt{4 - x^2}$, so one input (e.g. $x = 0$) yields two outputs ($+2$ and -2); the Vertical Line Test fails — the line $x = 0$ cuts the circle at two points.

Why the test is equivalent to the algebra: a vertical line $x = c$ collects all points of the graph having that single input c , so the number of intersections equals the number of outputs assigned to c . Two intersections is precisely the algebraic $\pm$ and precisely the violation of the definition of a function.

Q3. Find the inverse of the function $f(x) = 3x + 5$. [3 marks · Jun 2026 Q3(d)]

Step 1 — write $y = f(x)$. $y = 3x + 5$

Step 2 — solve for x in terms of y . $y - 5 = 3x \implies x = (y - 5)/3$

Step 3 — interchange the names of the variables to express the inverse as a function of x . $f^{-1}(x) = (x - 5)/3$

Verification. $f(f^{-1}(x)) = 3 \cdot (x - 5)/3 + 5 = (x - 5) + 5 = x \checkmark f^{-1}(f(x)) = ((3x + 5) - 5)/3 = 3x/3 = x \checkmark$

Answer: $f^{-1}(x) = (x - 5)/3$.

Condition that makes the inverse exist: $f(x) = 3x + 5$ is a linear function with non-zero slope 3, so it is one-to-one (injective) and onto $\mathbb{R}$ (surjective), i.e. a bijection from $\mathbb{R}$ to $\mathbb{R}$. Only a bijection has an inverse function, which is why this inverse exists and is itself a function.

Q4. A taxi service charges an extra penalty based on distance from central point. The penalty is modelled by function $f(x) = |x - 4|$. Where, x is the distance in 'km' from the city start marker. Find Domain, Range, X-intercept, Y-intercept. Draw and interpret the graph.

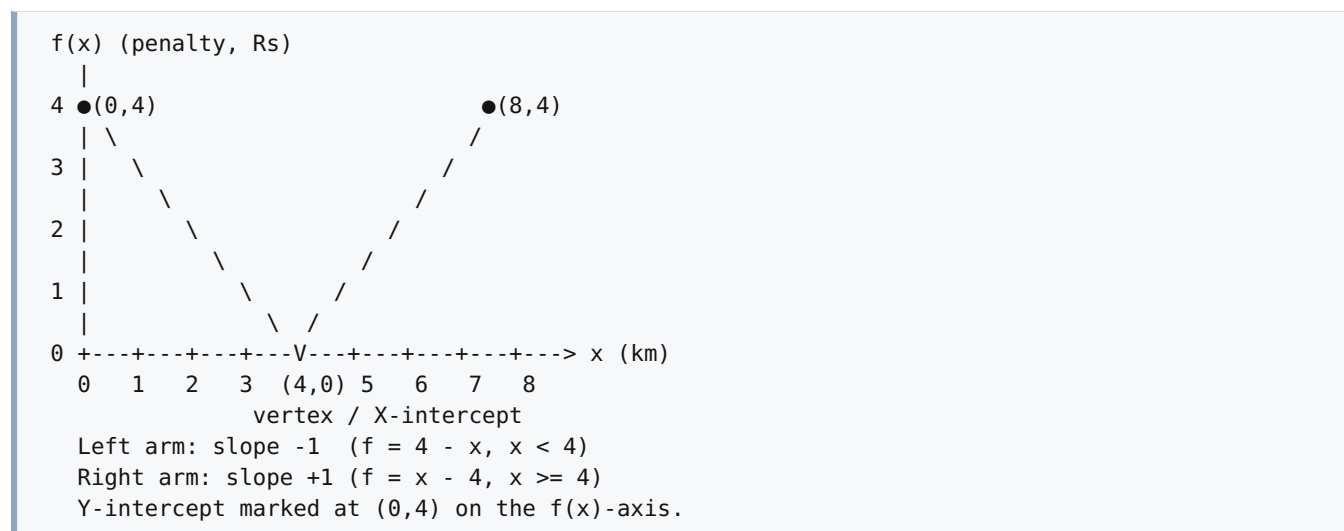
[8 marks · Jun 2026 Q4(a)]

[Reading adopted: the mathematical function is taken on all of $\mathbb{R}$, and the physical restriction $x \geq 0$ is stated separately, since a distance cannot be negative.]

Definition of the modulus used: $|x - 4| = x - 4$ when $x \geq 4$, and $|x - 4| = -(x - 4) = 4 - x$ when $x < 4$.

1. Domain. The expression $x - 4$ is defined for every real x and the modulus of any real number is defined, so no value is excluded. Domain = $\mathbb{R} = (-\infty, \infty)$. As a physical model of distance in km, the usable domain is $x \geq 0$, i.e. $[0, \infty)$.
2. Range. The modulus of a real number is never negative, and the value 0 is attained at $x = 4$. Range = $\{y : y \geq 0\} = [0, \infty)$.
3. X-intercept. Put $f(x) = 0$: $|x - 4| = 0 \rightarrow x - 4 = 0 \rightarrow x = 4$. X-intercept = $(4, 0)$. It is the only one.
4. Y-intercept. Put $x = 0$: $f(0) = |0 - 4| = |-4| = 4$. Y-intercept = $(0, 4)$.
5. Graph. A table of values:

x	0	1	2	3	4	5	6	7	8
$f(x) = x - 4 $	4	3	2	1	0	1	2	3	4



Answer: Domain = $\mathbb{I}$ (physically $x \geq 0$), Range = $[0, \infty)$, X-intercept = $(4, 0)$, Y-intercept = $(0, 4)$; the graph is a V with vertex at $(4, 0)$, left arm of slope -1 and right arm of slope $+1$.

Interpretation (the marks the question asks for explicitly): the V-shape says the penalty depends only on how far the trip point is from the 4 km mark, not on which side of it. The vertex at $(4, 0)$ means the penalty is zero exactly at 4 km from the city start marker — that is the "free" or ideal distance. Moving away from 4 km in either direction raises the penalty at the same constant rate

of ₹1 per km (slope magnitude 1), so a trip at 2 km and a trip at 6 km are penalised equally (₹2 each). The y-intercept (0, 4) says a trip starting right at the city marker, 0 km, carries the maximum penalty of ₹4 within the 0–4 km stretch. The range $[0, \infty)$ says the penalty can never be negative — the model never pays the customer back. The sharp corner at $x = 4$ says the rate of penalty switches abruptly from -1 to $+1$ there, which is why the function is continuous but not differentiable at $x = 4$.

Q5. $f(x) = x + 3$, $g(x) = 2x^2$. Find $(g \circ f)(2)$, $(f \circ g)(2)$, and state in one line why composition is not commutative. [practice]

$$(g \circ f)(2) = g(f(2)): f(2) = 2 + 3 = 5 \quad g(5) = 2 \times 5^2 = 2 \times 25 = 50$$

$$(f \circ g)(2) = f(g(2)): g(2) = 2 \times 2^2 = 2 \times 4 = 8 \quad f(8) = 8 + 3 = 11$$

Answer: $(g \circ f)(2) = 50$ and $(f \circ g)(2) = 11$.

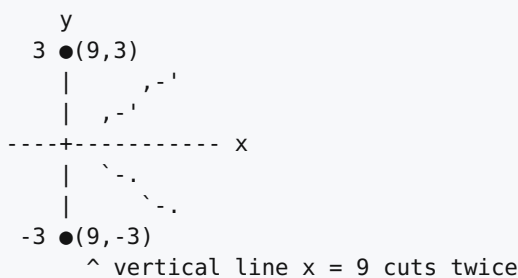
Why composition is not commutative: in $g \circ f$ the output of f is fed into g , while in $f \circ g$ the output of g is fed into f , so the two build different expressions — $(g \circ f)(x) = 2(x + 3)^2$ against $(f \circ g)(x) = 2x^2 + 3$ — and the values differ ($50 \neq 11$).

Q6. Show algebraically that $y^2 = x$ does not define y as a function of x , and say what the Vertical Line Test shows on its graph. [practice]

$$\text{Solve for } y: y^2 = x \rightarrow y = \pm\sqrt{x}$$

For any $x > 0$ there are two outputs. Take $x = 9$: $y = +3$ and $y = -3$, giving the points (9, 3) and (9, -3). One input with two outputs violates the definition of a function (exactly one output per input).

Vertical Line Test: the graph is a parabola opening to the right with vertex at the origin. The vertical line $x = 9$ meets it at (9, 3) and (9, -3) — two points — so the test fails.



Answer: $y^2 = x$ is not a function of x , since $y = \pm\sqrt{x}$ gives two outputs for each $x > 0$; the vertical line $x = 9$ cuts the graph at two points, so the Vertical Line Test fails.

Note on the restriction: if the output is restricted to $y \geq 0$, then $y = +\sqrt{x}$ alone is a function — the failure comes from keeping both signs, not from the equation being unusable.

Q7. Find the inverse of $f(x) = (2x - 7)/5$, and verify $f(f^{-1}(x)) = x$. [practice]

$$\text{Step 1 — set } y = f(x). \quad y = (2x - 7)/5$$

$$\text{Step 2 — solve for } x. \quad 5y = 2x - 7 \quad 2x = 5y + 7 \quad x = (5y + 7)/2$$

$$\text{Step 3 — interchange variable names. } f^{-1}(x) = (5x + 7)/2$$

$$\text{Verification. } f(f^{-1}(x)) = [2 \cdot (5x + 7)/2 - 7]/5 = [(5x + 7) - 7]/5 = 5x/5 = x \quad \checkmark \quad f^{-1}(f(x)) = [5 \cdot (2x - 7)/5 + 7]/2 = [(2x - 7) + 7]/2 = 2x/2 = x \quad \checkmark$$

Answer: $f^{-1}(x) = (5x + 7)/2$, and $f(f^{-1}(x)) = x$ is verified.

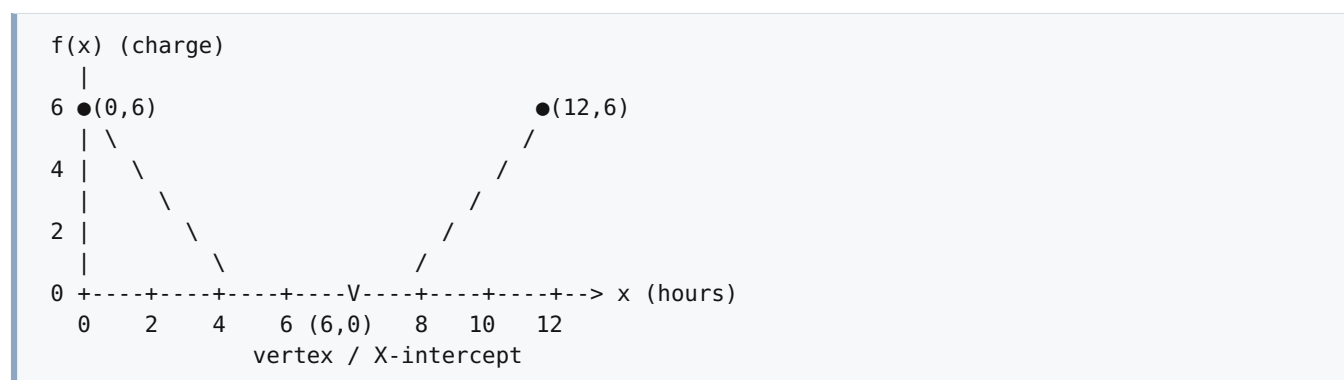
Why the inverse exists: f is linear with slope $2/5 \neq 0$, hence a bijection on $\mathbb{R}$; the two verification identities $f \circ f^{-1} = \text{identity}$ and $f^{-1} \circ f = \text{identity}$ are the definition of an inverse function, which is why writing both is the complete check.

Q8. A delay charge is modelled by $f(x) = |x - 6|$, x the hours after a scheduled time. Find domain, range, X-intercept, Y-intercept, draw the graph and interpret it. [practice]

Case split: $|x - 6| = x - 6$ for $x \geq 6$, and $6 - x$ for $x < 6$.

Domain = $\mathbb{R}$; as a model of hours after a scheduled time, $x \geq 0$, i.e. $[0, \infty)$. Range = $[0, \infty)$, since a modulus is never negative and the value 0 is reached at $x = 6$. X-intercept: $|x - 6| = 0 \rightarrow x = 6 \rightarrow (6, 0)$. Y-intercept: $f(0) = |-6| = 6 \rightarrow (0, 6)$.

x	0	2	4	6	8	10	12
f(x)	6	4	2	0	2	4	6



Answer: Domain = $\mathbb{I}$ (physically $x \geq 0$), Range = $[0, \infty)$, X-intercept = $(6, 0)$, Y-intercept = $(0, 6)$; V-shaped graph with vertex at $(6, 0)$.

Interpretation: the charge is zero exactly at 6 hours — the target time — and rises by 1 unit for every hour of deviation in either direction, so being 2 hours early ($x = 4$) and 2 hours late ($x = 8$) cost the same. The maximum charge within the 0–6 hour window is 6, at $x = 0$. The corner at $x = 6$ marks the instant where the rate of change flips from -1 to $+1$, so the model is continuous but not differentiable at $x = 6$.

Q9. $f(x) = |2x - 4| + 1$. Find the vertex, domain, range and both intercepts, and sketch it. [practice]

Step 1 — rewrite to standard V-form. $f(x) = |2x - 4| + 1 = |2(x - 2)| + 1 = 2|x - 2| + 1$

Step 2 — vertex. The modulus term is smallest ($= 0$) when $x - 2 = 0$, i.e. $x = 2$, and then $f(2) = 2 \times 0 + 1 = 1$. Vertex = $(2, 1)$.

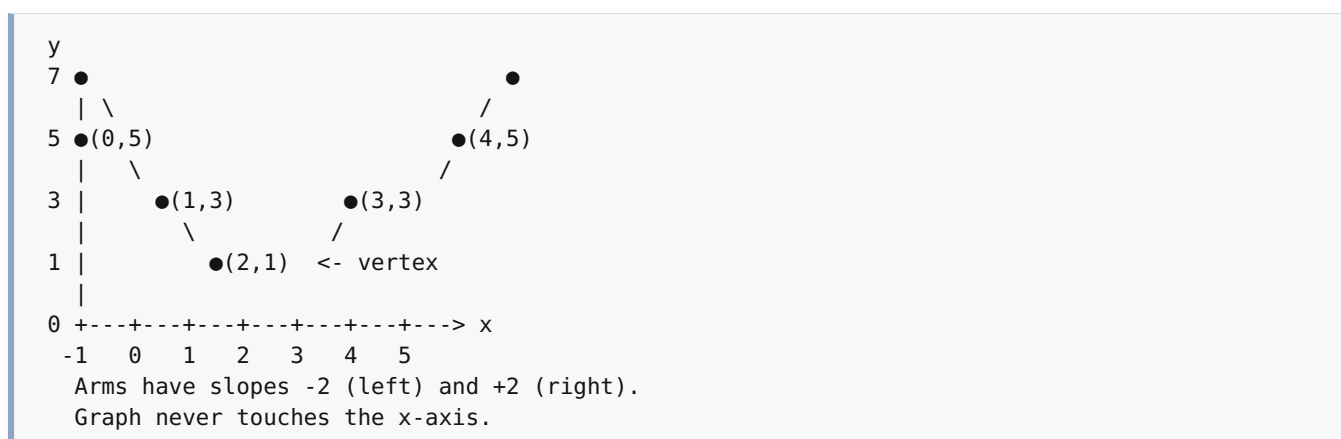
Step 3 — domain. Defined for every real x , so Domain = $\mathbb{R}$.

Step 4 — range. $2|x - 2| \geq 0$, so $f(x) \geq 1$, with equality at $x = 2$. Range = $[1, \infty) = \{y : y \geq 1\}$.

Step 5 — X-intercept. Put $f(x) = 0$: $2|x - 2| + 1 = 0 \rightarrow |x - 2| = -1/2$, which is impossible, because a modulus is never negative. There is no X-intercept — the graph lies entirely above the x -axis.

Step 6 — Y-intercept. Put $x = 0$: $f(0) = |2(0) - 4| + 1 = |-4| + 1 = 4 + 1 = 5$. Y-intercept = $(0, 5)$.

x	-1	0	1	2	3	4	5
f(x)	7	5	3	1	3	5	7



Answer: vertex (2, 1), Domain = $\mathbb{R}$, Range = $[1, \infty)$, no X-intercept, Y-intercept (0, 5).

Why there is no X-intercept, stated explicitly: the vertical shift +1 lifts the whole V so that its lowest point is at height $1 > 0$; since the minimum value of f is 1, $f(x) = 0$ has no solution. The factor 2 inside the modulus makes the arms twice as steep as those of $|x - 2|$, slopes -2 and $+2$, which is why the graph rises to 5 by the time x reaches 0 or 4.

Progressions

Q1. If the expressions $(5k + 1)$, $(6k + 5)$ and $(10k + 3)$ are three consecutive terms of an Arithmetic Progression (A.P.), determine the value of 'k'. Why is the relationship between these three terms sufficient to solve for the variable 'k'? [5 marks · Jun 2026 Q1(e)]

Condition used: if t_1 , t_2 , t_3 are consecutive terms of an A.P., the common difference is the same on both sides, so $t_2 - t_1 = t_3 - t_2$ (equivalently $2t_2 = t_1 + t_3$).

Substitution with $t_1 = 5k + 1$, $t_2 = 6k + 5$, $t_3 = 10k + 3$: $(6k + 5) - (5k + 1) = (10k + 3) - (6k + 5)$

Left side: $6k + 5 - 5k - 1 = k + 4$ Right side: $10k + 3 - 6k - 5 = 4k - 2$

So $k + 4 = 4k - 2$ $4 + 2 = 4k - k$ $6 = 3k$ $k = 2$

Check by substituting $k = 2$: $t_1 = 5(2) + 1 = 11$ $t_2 = 6(2) + 5 = 17$ $t_3 = 10(2) + 3 = 23$ Differences: $17 - 11 = 6$ and $23 - 17 = 6$. Equal, so 11, 17, 23 is an A.P. with $d = 6$. ✓

Answer: $k = 2$ (the three terms are 11, 17, 23 with common difference 6).

Answer to the rider: "consecutive terms of an A.P." is itself a mathematical condition, not just a description — it asserts the single equation $t_2 - t_1 = t_3 - t_2$. Substituting the three given expressions turns that one condition into one linear equation containing exactly one unknown, k , and one equation in one unknown has a unique solution. That is why no further data (no first term, no d , no n) is needed: the A.P. property supplies precisely the one equation required.

Q2. A company increases an employee's salary every year by ₹2,000. If the starting salary is ₹20,000 per month, find the salary in the 10th year. Also, find the total salary received in the first 10 years. [6 marks · Jun 2026 Q4(b)]

[Reading adopted: the yearly salaries form an A.P. in monthly terms, $a = ₹20,000$ per month and $d = ₹2,000$ per month; the 10th-year figure is asked as a monthly salary, and the 10-year total is given on both the monthly-figures basis and the actual annual-earnings basis.]

The monthly salary in successive years is 20,000, 22,000, 24,000, ... — an A.P. with $a = 20,000$, $d = 2,000$, $n = 10$.

Part 1 — salary in the 10th year. Formula: $t_n = a + (n - 1)d$ $t_{10} = 20,000 + (10 - 1) \times 2,000$ $t_{10} = 20,000 + 9 \times 2,000$ $t_{10} = 20,000 + 18,000$ $t_{10} = ₹38,000$ per month

Part 2 — total over the first 10 years. Formula: $S_n = n/2 [2a + (n - 1)d]$ $S_{10} = 10/2 [2 \times 20,000 + 9 \times 2,000]$ $S_{10} = 5 [40,000 + 18,000]$ $S_{10} = 5 \times 58,000$ $S_{10} = ₹2,90,000$

This ₹2,90,000 is the sum of the ten monthly-salary figures. The salary actually received over the 10 years, each year being 12 months, is Total earnings = $12 \times 2,90,000 = ₹34,80,000$

(Cross-check on S_{10} by the other form: $S_n = n/2 (a + t_n) = 10/2 \times (20,000 + 38,000) = 5 \times 58,000 = ₹2,90,000$ ✓)

Answer: salary in the 10th year = ₹38,000 per month; sum of the ten monthly-salary figures = ₹2,90,000, so the total salary received in the first 10 years = $12 \times 2,90,000 = ₹34,80,000$.

Statement of basis, which is where marks are lost: the question mixes a monthly figure (₹20,000 per month) with an annual increment (₹2,000 per year), so the basis must be declared. Here the A.P. is in monthly salary, hence t_{10} is a monthly amount; converting to actual money received requires the $\times 12$ factor, and both numbers are reported so that either expected basis is covered.

Q3. $(2k - 1)$, $(3k + 2)$ and $(6k - 1)$ are consecutive terms of an A.P. Find k , and state the condition you used. [practice]

Condition: $t_2 - t_1 = t_3 - t_2$ for consecutive A.P. terms.

$(3k + 2) - (2k - 1) = (6k - 1) - (3k + 2)$ Left: $3k + 2 - 2k + 1 = k + 3$ Right: $6k - 1 - 3k - 2 = 3k - 3$

$k + 3 = 3k - 3$ $3 + 3 = 3k - k$ $6 = 2k$ $k = 3$

Check: $t_1 = 2(3) - 1 = 5$, $t_2 = 3(3) + 2 = 11$, $t_3 = 6(3) - 1 = 17$. Differences 6 and 6. ✓

Answer: $k = 3$ (terms 5, 11, 17 with $d = 6$).

Condition stated: three terms are in A.P. iff the middle term is the arithmetic mean of its neighbours, $2t_2 = t_1 + t_3$, which is the same equation as $t_2 - t_1 = t_3 - t_2$ — one equation, one unknown, hence solvable.

Q4. The 4th term of an A.P. is 11 and the 9th is 26. Find a , d , the 20th term and the sum of the first 20 terms. [practice]

Formula used: $t_n = a + (n - 1)d$.

$t_4 = a + 3d = 11$... (i) $t_9 = a + 8d = 26$... (ii)

(ii) - (i): $5d = 15 \rightarrow d = 3$ From (i): $a + 3(3) = 11 \rightarrow a = 11 - 9 = 2$

$$20\text{th term: } t_{20} = a + 19d = 2 + 19 \times 3 = 2 + 57 = 59$$

$$\text{Sum: } S_{20} = n/2 [2a + (n - 1)d] = 20/2 [2 \times 2 + 19 \times 3] = 10 [4 + 57] = 10 \times 61 = 610$$

$$\text{Cross-check: } S_{20} = n/2 (a + t_{20}) = 10 \times (2 + 59) = 10 \times 61 = 610 \checkmark$$

Answer: $a = 2$, $d = 3$, $t_{20} = 59$, $S_{20} = 610$.

Why subtracting the two equations works: t_n is linear in n , so the difference of two given terms eliminates a and isolates d — $t_9 - t_4 = (9 - 4)d = 5d$. Two independent terms always determine an A.P. completely.

Q5. A machine's output falls by 150 units each year from 5,000 units. Find the output in the 8th year and the total output over 8 years. [practice]

Here $a = 5,000$ and the fall makes $d = -150$, $n = 8$.

$$\text{Output in the 8th year: } t_8 = a + 7d = 5,000 + 7 \times (-150) = 5,000 - 1,050 = 3,950 \text{ units}$$

$$\text{Total over 8 years: } S_8 = 8/2 [2 \times 5,000 + 7 \times (-150)] = 4 [10,000 - 1,050] = 4 \times 8,950 = 35,800 \text{ units}$$

$$\text{Cross-check: } S_8 = 8/2 (a + t_8) = 4 \times (5,000 + 3,950) = 4 \times 8,950 = 35,800 \checkmark$$

Answer: output in the 8th year = 3,950 units; total output over 8 years = 35,800 units.

Why d is negative: a decreasing sequence with a constant drop is still an A.P., with the common difference carrying a minus sign. Keeping the sign inside the formulas, rather than subtracting afterwards, is what keeps both results correct.

Q6. Find the sum of all multiples of 7 between 50 and 400. [practice]

Step 1 — identify the first and last terms. The first multiple of 7 greater than 50 is 56 (7×8); the last one less than 400 is 399 (7×57). So $a = 56$, $d = 7$, last term $l = 399$.

$$\text{Step 2 — number of terms. } l = a + (n - 1)d \quad 399 = 56 + (n - 1) \times 7 \quad 343 = (n - 1) \times 7 \quad n - 1 = 49 \quad n = 50$$

$$\text{Step 3 — sum. } S_n = n/2 (a + l) = 50/2 \times (56 + 399) = 25 \times 455 = 11,375$$

$$\text{Cross-check: the terms are } 7 \times 8 \text{ up to } 7 \times 57, \text{ so the sum is } 7 \times (8 + 9 + \dots + 57) = 7 \times [50/2 \times (8 + 57)] = 7 \times (25 \times 65) = 7 \times 1,625 = 11,375 \checkmark$$

Answer: 11,375.

Why the endpoints matter: "between 50 and 400" is read as strictly between, so 49 and 406 are outside the range and the series must start at 56 and stop at 399; mis-setting either endpoint changes n and so changes the sum.

Matrices, determinants and eigenvalues

Q1. Given the matrices A and B below, determine the values of x , y and z , if $A = B$. State the conditions when two matrices can be said equal. [5 marks · Jun 2026 Q1(f)]

$$A = \begin{bmatrix} 3 & x + y \\ x - y & 7 + z \end{bmatrix} \quad B = \begin{bmatrix} 3 & 6 \\ 8 & 4 \end{bmatrix}$$

Conditions for equality of two matrices (stated first, as the question asks): 1. They must be of the same order — the same number of rows and the same number of columns. 2. Every pair of corresponding elements must be equal, i.e. $a_{ij} = b_{ij}$ for all i and j . Both conditions must hold together; equality of order alone, or of some entries only, is not enough.

Here both A and B are 2×2 , so condition 1 is satisfied. Applying condition 2, element by element:

Position	A entry	B entry	Equation
(1,1)	3	3	$3 = 3$, identically true
(1,2)	$x + y$	6	$x + y = 6$
(2,1)	$x - y$	8	$x - y = 8$
(2,2)	$7 + z$	4	$7 + z = 4$

Solve $x + y = 6 \dots$ (i) and $x - y = 8 \dots$ (ii). (i) + (ii): $2x = 14 \rightarrow x = 7$ Substituting in (i): $7 + y = 6 \rightarrow y = 6 - 7 = -1$

Solve $7 + z = 4$: $z = 4 - 7 = -3$

Check by substituting back: $x + y = 7 + (-1) = 6 \checkmark$ $x - y = 7 - (-1) = 8 \checkmark$ $7 + z = 7 + (-3) = 4 \checkmark$

$$A = \begin{bmatrix} 3 & 6 \\ 8 & 4 \end{bmatrix} = B \quad \checkmark$$

Answer: $x = 7$, $y = -1$, $z = -3$.

Why this works: because matrix equality is defined entry by entry, a single matrix equation of order 2×2 is equivalent to four scalar equations. One of them is an identity ($3 = 3$), and the remaining three are exactly enough to determine the three unknowns x , y , z uniquely.

Q2. Without performing full expansion, determine the value of the determinant below. Which specific property of determinants justifies your result? [5 marks · Jun 2026 Q1(g)]

$$|A| = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 1 \\ 1 & 2 & 4 \end{vmatrix}$$

Examine the columns rather than expanding. $C_1 = (1, 2, 1)^T$ $C_2 = (2, 4, 2)^T$

Compare entry by entry: $2 = 2 \times 1$, $4 = 2 \times 2$, $2 = 2 \times 1$. Hence $C_2 = 2 C_1$, so the second column is a scalar multiple of the first — the two columns are proportional, i.e. linearly dependent.

Property applied: if any two rows or any two columns of a determinant are identical, or proportional (one a scalar multiple of the other), then the value of the determinant is zero. Equivalently, a determinant with linearly dependent rows or columns vanishes.

Therefore $|A| = 0$, without any expansion.

Answer: $|A| = 0$.

The property that justifies it, named explicitly: the determinant of a matrix having two proportional columns is zero — here $C_2 = 2C_1$. Consequently the rows/columns of A are linearly dependent, A is singular, and A^{-1} does not exist. (The underlying reason: factoring the 2 out of C_2 by the scalar-multiple property leaves a determinant with C_1 and C_2 identical, and a determinant with two identical columns changes sign when they are swapped while remaining the same determinant, so it must equal 0.)

Q3. Find the eigen values (λ_1, λ_2) of the square matrix A below. [5 marks · Jun 2026 Q1(h)]

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Definition used: the eigenvalues of A are the roots of the characteristic equation $|A - \lambda I| = 0$.

Form $A - \lambda I$:

$$A - \lambda I = \begin{bmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{bmatrix}$$

Set the determinant to zero: $(2 - \lambda)(2 - \lambda) - (1)(1) = 0$ $(2 - \lambda)^2 - 1 = 0$

Expand: $4 - 4\lambda + \lambda^2 - 1 = 0$ $\lambda^2 - 4\lambda + 3 = 0$

Factorise: $\lambda^2 - 3\lambda - \lambda + 3 = 0$ $\lambda(\lambda - 3) - 1(\lambda - 3) = 0$ $(\lambda - 1)(\lambda - 3) = 0$ $\lambda = 1$ or $\lambda = 3$

Answer: $\lambda_1 = 1$ and $\lambda_2 = 3$.

Verification against the invariants, which is the cheap confirming mark: the sum of the eigenvalues must equal the trace of A, and trace = $2 + 2 = 4$, while $\lambda_1 + \lambda_2 = 1 + 3 = 4$ ✓. The product of the eigenvalues must equal $|A| = (2)(2) - (1)(1) = 3$, while $\lambda_1 \times \lambda_2 = 1 \times 3 = 3$ ✓. Both eigenvalues are real and positive, as expected for a real symmetric matrix ($A = A^T$ here), and since neither is 0, A is non-singular.

Q4. A shopkeeper buys pens and notebooks, 2 pens and 1 notebook costs ₹50 whereas 1 pen and 2 notebooks costs ₹70. Find the cost of one pen and one notebook by using Cramer's rule.

[6 marks · Jun 2026 Q4(c)]

Let p = cost of one pen (₹) and n = cost of one notebook (₹).

Equations from the data: $2p + 1n = 50$... (i) $1p + 2n = 70$... (ii)

Matrix form:

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} p \\ n \end{bmatrix} = \begin{bmatrix} 50 \\ 70 \end{bmatrix}$$

Coefficient determinant D:

$$D = \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} = (2)(2) - (1)(1) = 4 - 1 = 3$$

$D = 3 \neq 0$, so by Cramer's rule the system has a unique solution.

D_p — replace the p-column (first column) by the constants:

$$D_p = \begin{vmatrix} 50 & 1 \\ 70 & 2 \end{vmatrix} = (50)(2) - (1)(70) = 100 - 70 = 30$$

D_n — replace the n-column (second column) by the constants:

$$D_n = \begin{vmatrix} 2 & 50 \\ 1 & 70 \end{vmatrix} = (2)(70) - (50)(1) = 140 - 50 = 90$$

Apply Cramer's rule: $p = D_p / D = 30 / 3 = 10$ $n = D_n / D = 90 / 3 = 30$

Check in both original equations: (i) $2(10) + 30 = 20 + 30 = 50$ ✓ (ii) $10 + 2(30) = 10 + 60 = 70$ ✓

Answer: cost of one pen = ₹10 and cost of one notebook = ₹30.

Condition that must be stated: Cramer's rule applies only when the coefficient determinant $D \neq 0$. Here $D = 3 \neq 0$, so the coefficient matrix is non-singular and the system is consistent with exactly one solution. Had D been 0, the rule would fail and the system would have either no solution or infinitely many.

Q5. If $A = \begin{bmatrix} 3 & 5 \\ 2 & 4 \end{bmatrix}$, show that $\frac{1}{2}(A + A^T)$ is symmetric. [5 marks · Jun 2026 Q5(d)]

$$A = \begin{bmatrix} 3 & 5 \\ 2 & 4 \end{bmatrix}$$

Step 1 — write the transpose (rows of A become columns of A^T).

$$A^T = \begin{bmatrix} 3 & 2 \\ 5 & 4 \end{bmatrix}$$

Step 2 — add, entry by entry.

$$A + A^T = \begin{bmatrix} 3+3 & 5+2 \\ 2+5 & 4+4 \end{bmatrix} = \begin{bmatrix} 6 & 7 \\ 7 & 8 \end{bmatrix}$$

Step 3 — multiply by $\frac{1}{2}$ (scalar multiplication multiplies every entry). Call the result B .

$$B = (1/2)(A + A^T) = \begin{bmatrix} 3 & 7/2 \\ 7/2 & 4 \end{bmatrix}$$

Step 4 — definition of symmetric: a square matrix B is symmetric iff $B^T = B$, i.e. $b_{ij} = b_{ji}$ for all i, j .

Take the transpose of B by interchanging rows and columns:

$$B^T = \begin{bmatrix} 3 & 7/2 \\ 7/2 & 4 \end{bmatrix}$$

Compare entry by entry with B : $b_{11} = 3 = b_{11}$, $b_{12} = 7/2 = b_{21}$, $b_{21} = 7/2 = b_{12}$, $b_{22} = 4 = b_{22}$. Every corresponding entry agrees, and both are 2×2 , so by the conditions of matrix equality

$B^t = B$, hence $B = \frac{1}{2}(A + A^T)$ is symmetric.

Answer: $\frac{1}{2}(A + A^S) = \begin{bmatrix} 3 & 7/2 \\ 7/2 & 4 \end{bmatrix}$, and since its transpose equals itself, it is symmetric. Proved.

The general reason, worth stating: using $(X + Y)^T = X^T + Y^T$, $(X^T)^T = X$ and $(kX)^T = kX^T$, $[\frac{1}{2}(A + A^T)]^T = \frac{1}{2}(A^T + (A^T)^T) = \frac{1}{2}(A^T + A) = \frac{1}{2}(A + A^T)$, so $\frac{1}{2}(A + A^T)$ is symmetric for every square matrix

A, not just this one. Note also that the proof is incomplete until the line " $B^T = B$, hence B is symmetric" is actually written.

Q6. $A = \begin{bmatrix} 5 & p+q \\ p-q & 2+r \end{bmatrix}$, $B = \begin{bmatrix} 5 & 9 \\ 3 & 7 \end{bmatrix}$. Find p , q , r if $A = B$, and state the conditions for matrix equality. [practice]

Conditions for matrix equality: (i) same order, (ii) all corresponding elements equal.

Both matrices are 2×2 , so condition (i) holds. Equating corresponding entries: (1,1): $5 = 5$, identically true (1,2): $p + q = 9$... (i) (2,1): $p - q = 3$... (ii) (2,2): $2 + r = 7$... (iii)

(i) + (ii): $2p = 12 \rightarrow p = 6$ From (i): $6 + q = 9 \rightarrow q = 3$ From (iii): $r = 7 - 2 = 5$

Check: $p + q = 6 + 3 = 9 \checkmark$, $p - q = 6 - 3 = 3 \checkmark$, $2 + r = 2 + 5 = 7 \checkmark$

Answer: $p = 6$, $q = 3$, $r = 5$.

Reason the system is solvable: entry-wise equality of a 2×2 matrix equation gives four scalar equations; one is an identity and the other three determine the three unknowns uniquely.

Q7. Without expanding, evaluate the determinant $\begin{vmatrix} 2 & 4 & 6 \\ 1 & 3 & 5 \\ 2 & 4 & 6 \end{vmatrix}$ and name the property used. [practice]

$$|A| = \begin{vmatrix} 2 & 4 & 6 \\ 1 & 3 & 5 \\ 2 & 4 & 6 \end{vmatrix}$$

Observe $R_1 = (2, 4, 6)$ and $R_3 = (2, 4, 6)$. The first and third rows are identical.

Property: if any two rows (or columns) of a determinant are identical, the determinant is zero.

Answer: $|A| = 0$.

Property named: two identical rows ($R_1 = R_3$) make the determinant vanish. The reason behind the property is that interchanging two rows multiplies a determinant by -1 ; swapping the two identical rows leaves the determinant unchanged, so $|A| = -|A|$, forcing $|A| = 0$. The matrix is therefore singular.

Q8. Without expanding, evaluate the determinant $\begin{vmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 4 & 5 & 6 \end{vmatrix}$ and name the property used. [practice]

$$|A| = \begin{vmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 4 & 5 & 6 \end{vmatrix}$$

The second row consists entirely of zeros.

Property: if every element of any one row (or column) of a determinant is zero, the determinant is zero.

Answer: $|A| = 0$.

Property named: a zero row makes the determinant vanish. The reason: expanding along that row gives a sum in which every term carries a factor 0, so the total is 0 — which is why no expansion needs to be written out. The matrix is singular.

Q9. Find the eigenvalues of $\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ and of $\begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}$. [practice]

Matrix 1:

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

$$|A - \lambda I| = 0: (4 - \lambda)(3 - \lambda) - (1)(2) = 0 \quad 12 - 4\lambda - 3\lambda + \lambda^2 - 2 = 0 \quad \lambda^2 - 7\lambda + 10 = 0 \quad (\lambda - 2)(\lambda - 5) = 0 \quad \lambda_1 = 2, \lambda_2 = 5$$

$$\text{Check: trace} = 4 + 3 = 7 = 2 + 5 \checkmark; |A| = 12 - 2 = 10 = 2 \times 5 \checkmark$$

Matrix 2:

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}$$

$$|A - \lambda I| = (3 - \lambda)(5 - \lambda) - 0 = 0 \rightarrow \lambda = 3 \text{ or } \lambda = 5$$

$$\text{Check: trace} = 8 = 3 + 5 \checkmark; |A| = 15 = 3 \times 5 \checkmark$$

Answer: eigenvalues of $\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ are 2 and 5; eigenvalues of $\begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}$ are 3 and 5.

Property worth noting: for a diagonal (or triangular) matrix the characteristic determinant is already a product of the diagonal entries minus λ , so the eigenvalues are just the diagonal entries themselves — no expansion is needed for the second matrix. For any 2×2 matrix the characteristic equation can be written directly as $\lambda^2 - (\text{trace})\lambda + (\det) = 0$, which is the shortcut used as the check above.

Q10. $3x + 2y = 12$ and $x + 4y = 14$. Solve by Cramer's rule, showing D , D_x and D_y . [practice]

$$\begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 12 \\ 14 \end{bmatrix}$$

$$D = (3)(4) - (2)(1) = 12 - 2 = 10. \text{ Since } D = 10 \neq 0, \text{ a unique solution exists.}$$

$$D_x = \begin{vmatrix} 12 & 2 \\ 14 & 4 \end{vmatrix} = (12)(4) - (2)(14) = 48 - 28 = 20$$
$$D_y = \begin{vmatrix} 3 & 12 \\ 1 & 14 \end{vmatrix} = (3)(14) - (12)(1) = 42 - 12 = 30$$

$$x = D_x/D = 20/10 = 2 \quad y = D_y/D = 30/10 = 3$$

$$\text{Check: } 3(2) + 2(3) = 6 + 6 = 12 \checkmark; 2 + 4(3) = 2 + 12 = 14 \checkmark$$

Answer: $x = 2, y = 3$.

Condition stated: Cramer's rule requires $D \neq 0$; here $D = 10$, so the coefficient matrix is non-singular and the solution is unique. Each numerator determinant is formed by replacing the column of the variable being solved for with the column of constants.

Q11. $A = \begin{bmatrix} 1 & 4 \\ 6 & 2 \end{bmatrix}$. Show $\frac{1}{2}(A + A^T)$ is symmetric and $\frac{1}{2}(A - A^T)$ is skew-symmetric, and hence write A as their sum. [practice]

$$A = \begin{bmatrix} 1 & 4 \\ 6 & 2 \end{bmatrix} \quad A^T = \begin{bmatrix} 1 & 6 \\ 4 & 2 \end{bmatrix}$$

Symmetric part $P = \frac{1}{2}(A + A^T)$:

$$A + A^T = \begin{bmatrix} 2 & 10 \\ 10 & 4 \end{bmatrix} \Rightarrow P = \begin{bmatrix} 1 & 5 \\ 5 & 2 \end{bmatrix}$$

P^T : interchanging rows and columns gives $\begin{bmatrix} 1 & 5 \\ 5 & 2 \end{bmatrix} = P$. Since $P^T = P$, P is symmetric. ✓

Skew-symmetric part $Q = \frac{1}{2}(A - A^T)$:

$$A - A^T = \begin{bmatrix} 1-1 & 4-6 \\ 6-4 & 2-2 \end{bmatrix} = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix} \Rightarrow Q = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$Q^T = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ and $-Q = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. Since $Q^T = -Q$ (and the diagonal entries are 0, as a skew-symmetric matrix requires), Q is skew-symmetric. ✓

Sum:

$$P + Q = \begin{bmatrix} 1 & 5 \\ 5 & 2 \end{bmatrix} + \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1+0 & 5-1 \\ 5+1 & 2+0 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 6 & 2 \end{bmatrix} = A \quad \checkmark$$

Answer: $P = \frac{1}{2}(A + A^T) = \begin{bmatrix} 1 & 5 \\ 5 & 2 \end{bmatrix}$ is symmetric, $Q = \frac{1}{2}(A - A^T) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ is skew-symmetric, and $A = P + Q$.

The general theorem demonstrated: every square matrix A can be written uniquely as the sum of a symmetric and a skew-symmetric matrix, $A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$, because $P^T = \frac{1}{2}(A^T + A) = P$ and $Q^T = \frac{1}{2}(A^T - A) = -Q$ identically. Each proof must end with the explicit line $P^T = P$ and $Q^T = -Q$.

Linear spaces

Q1. Suppose three resources x , y and z in a system satisfy the condition $x + y - z = 0$. Find a basis for the subspace of all vectors (x, y, z) satisfying this condition. [5 marks · Jun 2026 Q5(a)]

Let $W = \{(x, y, z) \in \mathbb{R}^3 : x + y - z = 0\}$.

Step 1 — solve the constraint for one variable. $x + y - z = 0 \rightarrow z = x + y$

So x and y may be chosen freely and z is then determined.

Step 2 — write the general vector of W . Any member of W has the form $(x, y, z) = (x, y, x + y)$

Step 3 — split the general vector along the free parameters. $(x, y, x + y) = (x, 0, x) + (0, y, y) = x(1, 0, 1) + y(0, 1, 1)$

So every vector of W is a linear combination of $v_1 = (1, 0, 1)$ and $v_2 = (0, 1, 1)$, i.e. $W = \text{span}\{v_1, v_2\}$.

Step 4 — check these two vectors lie in W . For v_1 : $1 + 0 - 1 = 0$ ✓ For v_2 : $0 + 1 - 1 = 0$ ✓

Step 5 — check linear independence. Suppose $a(1, 0, 1) + b(0, 1, 1) = (0, 0, 0)$. Comparing components: first component: $a = 0$ second component: $b = 0$ Only the trivial combination gives the zero vector, so v_1 and v_2 are linearly independent. (Equivalently, neither is a scalar multiple of the other.)

Answer: a basis is $\{(1, 0, 1), (0, 1, 1)\}$, and $\dim W = 2$.

Why this is a basis, stated explicitly: a basis of a subspace is a linearly independent set that spans it, and both requirements have been verified — Step 3 shows spanning and Step 5 shows independence. Geometrically W is the plane $x + y - z = 0$ through the origin in $\mathbb{R}^3$, a 2-dimensional subspace; this agrees with the counting rule that one independent linear condition on $\mathbb{R}^3$ leaves $3 - 1 = 2$ free parameters, hence dimension 2. W is genuinely a subspace because it contains the zero vector ($0 + 0 - 0 = 0$) and is closed under addition and scalar multiplication, the constraint being homogeneous (right-hand side 0).

Q2. Find a basis for the subspace of all (x, y, z) in $\mathbb{R}^3$ satisfying $2x - y + z = 0$, and state its dimension. [practice]

Solve the constraint for z : $2x - y + z = 0 \rightarrow z = y - 2x$

General vector: $(x, y, z) = (x, y, y - 2x) = (x, 0, -2x) + (0, y, y) = x(1, 0, -2) + y(0, 1, 1)$

Check both lie in the subspace: $(1, 0, -2)$: $2(1) - 0 + (-2) = 0$ ✓ $(0, 1, 1)$: $2(0) - 1 + 1 = 0$ ✓

Independence: $a(1, 0, -2) + b(0, 1, 1) = (0, 0, 0)$ gives $a = 0$ from the first component and $b = 0$ from the second, so only the trivial combination works — independent.

Answer: basis $\{(1, 0, -2), (0, 1, 1)\}$, dimension = 2.

Why dimension 2: x and y are free parameters while z is forced by the single homogeneous equation, so $3 - 1 = 2$ degrees of freedom remain; the set is a plane through the origin in $\mathbb{R}^3$.

Q3. Find a basis for the subspace of all (x, y, z) with $x = y$, and state its dimension. [practice]

The constraint $x = y$, i.e. $x - y = 0$, makes y equal to x and leaves z free.

General vector: $(x, y, z) = (x, x, z) = (x, x, 0) + (0, 0, z) = x(1, 1, 0) + z(0, 0, 1)$

Both generators satisfy $x = y$: for $(1, 1, 0)$, $1 = 1$ ✓; for $(0, 0, 1)$, $0 = 0$ ✓.

Independence: $a(1, 1, 0) + b(0, 0, 1) = (0, 0, 0)$ gives $a = 0$ (first component) and $b = 0$ (third component) — independent.

Answer: basis $\{(1, 1, 0), (0, 0, 1)\}$, dimension = 2.

Why it is a subspace of dimension 2: the condition $x - y = 0$ is a single homogeneous linear equation, so the solution set contains the origin, is closed under addition and scalar multiples, and has $3 - 1 = 2$ free parameters, namely x and z .

Q4. Show that $\{(1, 0, 1), (0, 1, 1)\}$ is linearly independent and name the subspace it spans. [practice]

Independence test: solve $a(1, 0, 1) + b(0, 1, 1) = (0, 0, 0)$. Component-wise: 1st: $a + 0 = 0 \rightarrow a = 0$
2nd: $0 + b = 0 \rightarrow b = 0$ 3rd: $a + b = 0$, consistent with $a = b = 0$

Only the trivial solution $a = b = 0$ exists, hence the set is linearly independent.

The subspace spanned. A general combination is $a(1, 0, 1) + b(0, 1, 1) = (a, b, a + b)$ Writing $x = a$, $y = b$, $z = a + b$ gives $z = x + y$, i.e. $x + y - z = 0$.

Answer: the set is linearly independent, and it spans the plane $W = \{(x, y, z) : x + y - z = 0\}$ in $\mathbb{R}^3$, a 2-dimensional subspace.

Why these two facts together matter: a linearly independent set that spans W is by definition a basis of W , so $\{(1, 0, 1), (0, 1, 1)\}$ is a basis of that plane and $\dim W = 2$. Independence can also be seen at a glance here because neither vector is a scalar multiple of the other.

Counting and the binomial theorem

Q1. State Binomial theorem and use it to derive formula for identities $(a + b)^2$, and $(a + b)^3$.

[6 marks · Jun 2026 Q3(a)]

Statement of the binomial theorem. For any real (or complex) a , b and any positive integer n ,

$$(a + b)^n = \sum_{r=0}^n C(n, r) a^{n-r} b^r = C(n, 0)a^n + C(n, 1)a^{n-1}b + C(n, 2)a^{n-2}b^2 + \dots + C(n, n)b^n$$

where the binomial coefficient is $C(n, r) = n! / [r! (n - r)!]$, with $0! = 1$.

Derivation of $(a + b)^2$. Put $n = 2$ in the theorem; r runs 0, 1, 2:

$$(a + b)^2 = C(2, 0) a^2 b^0 + C(2, 1) a^1 b^1 + C(2, 2) a^0 b^2$$

Coefficients from the formula: $C(2, 0) = 2! / (0! 2!) = 2/2 = 1$ $C(2, 1) = 2! / (1! 1!) = 2/1 = 2$ $C(2, 2) = 2! / (2! 0!) = 2/2 = 1$

Substituting: $(a + b)^2 = 1 \cdot a^2 + 2 \cdot ab + 1 \cdot b^2$ **$(a + b)^2 = a^2 + 2ab + b^2$**

Derivation of $(a + b)^3$. Put $n = 3$; r runs 0, 1, 2, 3:

$$(a + b)^3 = C(3, 0) a^3 + C(3, 1) a^2 b + C(3, 2) ab^2 + C(3, 3) b^3$$

Coefficients: $C(3, 0) = 3! / (0! 3!) = 6/6 = 1$ $C(3, 1) = 3! / (1! 2!) = 6/2 = 3$ $C(3, 2) = 3! / (2! 1!) = 6/2 = 3$ $C(3, 3) = 3! / (3! 0!) = 6/6 = 1$

Substituting: $(a + b)^3 = 1 \cdot a^3 + 3 \cdot a^2 b + 3 \cdot ab^2 + 1 \cdot b^3$ **$(a + b)^3 = a^3 + 3a^2 b + 3ab^2 + b^3$**

Answer: $(a + b)^n = \sum_{r=0}^n C(n, r) a^{n-r} b^r$; with $n = 2$ it gives $a^2 + 2ab + b^2$, and with $n = 4$ terms at $n = 3$ it gives $a^3 + 3a^2 b + 3ab^2 + b^3$.

Why the derivation, not the memorised identity, earns the marks: each coefficient must be shown to come out of $C(n, r)$, and each term's exponents must be shown to satisfy (power of a) + (power of b) = n . Note also the structural facts the theorem gives free: the expansion of $(a + b)^n$ has $n + 1$ terms, the coefficients are symmetric because $C(n, r) = C(n, n - r)$ (hence 1, 2, 1 and 1, 3, 3, 1 — the rows of Pascal's triangle), and the sum of all coefficients is 2^n , which checks as $1 + 2 + 1 = 4 = 2^2$ and $1 + 3 + 3 + 1 = 8 = 2^3$.

Q2. A company counts to create an access code using two different letters from the English alphabets followed by three different digits from 0 to 9. How many such access codes can be formed? [5 marks · Jun 2026 Q5(b)]

[Reading adopted: 26 letters, case ignored; "different" means no repetition within the letters and no repetition within the digits; position matters, so each stage is a permutation.]

Structure of the code: $L_1 L_2 D_1 D_2 D_3$ — two letter positions followed by three digit positions.

Stage 1 — the two letters, no repetition: first letter: 26 choices second letter: 25 choices (the letter already used is barred) letter arrangements = $26 \times 25 = 650$ (Equivalently the permutation ${}^{26}P_2 = 26!/24! = 650$.)

Stage 2 — the three digits from 0-9, no repetition: first digit: 10 choices second digit: 9 choices third digit: 8 choices digit arrangements = $10 \times 9 \times 8 = 720$ (Equivalently ${}^{10}P_3 = 10!/7! = 720$.)

Stage 3 — combine by the fundamental principle of counting (multiplication rule), the two stages being independent: total codes = 650×720

$$650 \times 720 = 650 \times 700 + 650 \times 20 = 455,000 + 13,000 = 468,000$$

Answer: 468,000 access codes can be formed.

Justification of the method: the multiplication principle applies because the choice of letters places no restriction on the choice of digits, so the counts multiply rather than add. Permutations and not combinations are used because the code's positions are ordered — AB7 and BA7 are different codes — and the counts fall by one at each step ($26 \rightarrow 25$, $10 \rightarrow 9 \rightarrow 8$) precisely because "different" bars repetition. For contrast, if repetition were allowed the answer would be $26^2 \times 10^3 = 676 \times 1,000 = 676,000$.

Q3. State the binomial theorem and use it to expand $(a + b)^4$ and $(2x - 3)^3$. [practice]

Theorem: $(a + b)^n = \sum_{r=0}^n C(n, r) a^{n-r} b^r$, with $C(n, r) = n!/[r!(n - r)!]$.

Expansion of $(a + b)^4$. Put $n = 4$; coefficients: $C(4,0) = 1$, $C(4,1) = 4$, $C(4,2) = 4!/(2!2!) = 24/4 = 6$, $C(4,3) = 4$, $C(4,4) = 1$

$$(a + b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$$

Expansion of $(2x - 3)^3$. Apply the theorem with $a = 2x$, $b = -3$, $n = 3$:

$$(2x - 3)^3 = C(3,0)(2x)^3 + C(3,1)(2x)^2(-3) + C(3,2)(2x)(-3)^2 + C(3,3)(-3)^3 = 1 \times 8x^3 + 3 \times 4x^2 \times (-3) + 3 \times 2x \times 9 + 1 \times (-27) = 8x^3 - 36x^2 + 54x - 27$$

Check at $x = 1$: $(2 - 3)^3 = (-1)^3 = -1$, and $8 - 36 + 54 - 27 = -1 \checkmark$

Answer: $(a + b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$; $(2x - 3)^3 = 8x^3 - 36x^2 + 54x - 27$.

Point to state when a term is negative: taking $b = -3$ makes the signs alternate automatically, because $b^r = (-3)^r$ is negative for odd r and positive for even r — no separate sign rule is needed. The coefficient of each term must include the power of the whole bracket contents, e.g. $(2x)^2 = 4x^2$, not $2x^2$.

Q4. Find the coefficient of x^3 in the expansion of $(1 + 2x)^5$. [practice]

General term of $(1 + 2x)^5$ by the binomial theorem, with $a = 1$, $b = 2x$, $n = 5$: $T_{r+1} = C(5, r) (1)^{5-r} (2x)^r = C(5, r) 2^r x^r$

The term in x^3 needs $r = 3$: $C(5, 3) 2^3 x^3$

$$C(5, 3) = 5!/(3! 2!) = 120/(6 \times 2) = 10 \quad 2^3 = 8 \quad \text{coefficient} = 10 \times 8 = 80$$

Answer: the coefficient of x^3 is 80 (the term being $80x^3$).

Why the general term is used rather than the full expansion: the power of x in T_{r+1} equals r , so matching r to the required power picks out the one needed term directly. Note that the factor 2 must be raised to the same power r as x , which is why the coefficient is $C(5,3) \times 2^3$ and not just $C(5,3)$.

Q5. A code is one letter followed by four different digits. How many codes are possible? How many if the digits may repeat? [practice]

Case 1 — four different digits. letter: 26 choices digits, no repetition: $10 \times 9 \times 8 \times 7 = 5,040$ total = $26 \times 5,040 = 1,31,040$

Case 2 — digits may repeat. letter: 26 choices digits, repetition allowed: $10 \times 10 \times 10 \times 10 = 10^4 = 10,000$ total = $26 \times 10,000 = 2,60,000$

Answer: 1,31,040 codes with four different digits, and 2,60,000 codes if the digits may repeat.

Which principle is used and why the counts differ: the multiplication rule applies in both cases because the stages are independent. With "different" digits each successive position loses one option (10, 9, 8, 7), making the count the permutation ${}^{10}P_4 = 5,040$; with repetition allowed every position has the full 10 options, giving 10^4 . Repetition therefore almost doubles the code space here.

Q6. How many 4-digit numbers with no repeated digit can be formed from 1–9? How many of those are even? [practice]

Part 1 — all such numbers. Digits available: 1, 2, ..., 9 (nine digits; 0 is not available, so no leading-zero issue). No repetition, order matters: 1st place: 9 choices 2nd place: 8 3rd place: 7 4th place: 6 total = $9 \times 8 \times 7 \times 6 = 3,024$ ($= {}^9P_4$)

Part 2 — the even ones. A number is even iff its units digit is even. Among 1–9 the even digits are 2, 4, 6, 8 — four options. Fix the units place first (the restricted position): units place: 4 choices remaining three places, from the 8 unused digits, no repetition: $8 \times 7 \times 6 = 336$ total even = $4 \times 336 = 1,344$

Cross-check by symmetry: the odd digits are 1, 3, 5, 7, 9 — five options — giving $5 \times 336 = 1,680$ odd numbers, and $1,344 + 1,680 = 3,024$ ✓

Answer: 3,024 such 4-digit numbers, of which 1,344 are even.

Why the restricted position is filled first: the evenness condition constrains only the units digit, so choosing it before the free positions keeps the remaining counts uniform ($8 \times 7 \times 6$ whichever even digit was taken). Filling the places in the natural left-to-right order would make the number of options for the last place depend on earlier choices, forcing a case split.

Q7. In how many ways can a committee of 3 be chosen from 8 people? And if one named person must be included? [practice]

Part 1. A committee is an unordered selection, so use combinations: $C(8, 3) = 8!/(3! 5!) = (8 \times 7 \times 6)/(3 \times 2 \times 1) = 336/6 = 56$

Part 2. If a particular person must be on the committee, that one seat is fixed; the remaining 2 members are chosen from the other 7 people: $C(7, 2) = 7!/(2! 5!) = (7 \times 6)/2 = 42/2 = 21$

Answer: 56 committees in all, of which 21 contain the named person.

Why combinations and not permutations: a committee has no internal order — choosing {A, B, C} is the same committee however the three names are listed — so each selection must be counted once, which is $C(8,3)$ and not ${}^8P_3 = 336$. For the second part, forcing one member reduces both the pool ($8 \rightarrow 7$) and the number still to be chosen ($3 \rightarrow 2$). As a check, the committees excluding that person number $C(7, 3) = 35$, and $21 + 35 = 56$ ✓

Limits, continuity, derivatives and integration

Q1. Verify that the limit $\lim_{(x \rightarrow 4)} |x - 4|/(x - 4)$ exists or not. Support your verification statement with suitable arguments. [5 marks · Jun 2026 Q2(a)]

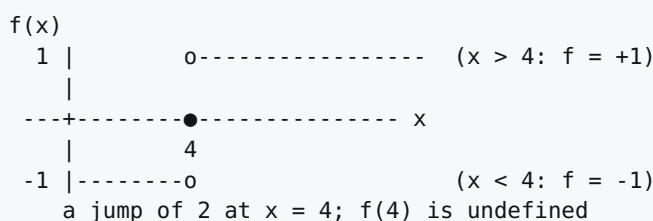
Test to be applied, stated first: $\lim_{(x \rightarrow a)} f(x)$ exists iff the left-hand limit and the right-hand limit both exist, are finite, and are equal: $\text{LHL} = \lim_{(x \rightarrow a^-)} f(x) = \lim_{(x \rightarrow a^+)} f(x) = \text{RHL}$.

Definition of the modulus: $|x - 4| = x - 4$ when $x - 4 \geq 0$, i.e. $x \geq 4$ $|x - 4| = -(x - 4)$ when $x - 4 < 0$, i.e. $x < 4$

Right-hand limit ($x \rightarrow 4^+$, so $x > 4$ and $|x - 4| = x - 4$): $\text{RHL} = \lim_{(x \rightarrow 4^+)} (x - 4)/(x - 4) = \lim_{(x \rightarrow 4^+)} 1 = 1$

Left-hand limit ($x \rightarrow 4^-$, so $x < 4$ and $|x - 4| = -(x - 4)$): $\text{LHL} = \lim_{(x \rightarrow 4^-)} -(x - 4)/(x - 4) = \lim_{(x \rightarrow 4^-)} (-1) = -1$

Comparison: $\text{LHL} = -1$, $\text{RHL} = +1$, and $-1 \neq +1$.



Answer: the limit $\lim_{(x \rightarrow 4)} |x - 4|/(x - 4)$ does not exist.

Supporting argument, which is where the marks sit: the function equals the constant -1 everywhere to the left of 4 and the constant $+1$ everywhere to the right of 4, so it approaches two different values from the two sides. The existence criterion requires a single value to be approached irrespective of the direction, i.e. $\text{LHL} = \text{RHL}$; since the two one-sided limits are finite but unequal ($-1 \neq 1$), the two-sided limit fails to exist. The function has a jump discontinuity of size 2 at $x = 4$, and note separately that $f(4) = 0/0$ is undefined — but that alone would not prevent a limit from existing; it is the inequality of the one-sided limits that does.

Q2. Using the First Principle (Delta Method), find the derivative of the function $f(x) = x^2$. [5 marks · Jun 2026 Q2(b)]

Definition written first (first principle / delta method): $f'(x) = \lim_{(h \rightarrow 0)} [f(x + h) - f(x)] / h$

Step 1 — form $f(x + h)$. $f(x) = x^2$, so $f(x + h) = (x + h)^2 = x^2 + 2xh + h^2$

Step 2 — form the difference. $f(x + h) - f(x) = (x^2 + 2xh + h^2) - x^2 = 2xh + h^2$

Step 3 — divide by h (legitimate since $h \neq 0$ while the limit is being taken). $[f(x + h) - f(x)]/h = (2xh + h^2)/h = h(2x + h)/h = 2x + h$

Step 4 — take the limit as $h \rightarrow 0$. $f'(x) = \lim_{(h \rightarrow 0)} (2x + h) = 2x + 0 = 2x$

Answer: $f'(x) = 2x$.

Why each step is needed: the quotient $[f(x+h) - f(x)]/h$ is the slope of the chord (secant) joining the points $(x, f(x))$ and $(x + h, f(x + h))$, and the derivative is defined as the limiting slope of that chord as the second point slides into the first, i.e. the slope of the tangent. The cancellation of h in Step 3 is essential — before it, substituting $h = 0$ gives the indeterminate form $0/0$; after it, the

expression $2x + h$ is continuous in h and the limit can be read off by direct substitution. The result agrees with the standard power rule $d(x^n)/dx = n x^{(n-1)}$ with $n = 2$.

Q3. Evaluate the integral $\int x^3 dx$. Explain why the answer to the integral will be incomplete if the constant (C) is not included in the result. [5 marks · Jun 2026 Q2(c)]

Formula used (power rule for integration): $\int x^n dx = x^{(n+1)}/(n + 1) + C$, valid for $n \neq -1$

Substituting $n = 3$: $\int x^3 dx = x^{(3+1)}/(3 + 1) + C = x^4/4 + C$

Verification by differentiation: $d/dx [x^4/4 + C] = (1/4) \times 4x^3 + 0 = x^3 \checkmark$

Answer: $\int x^3 dx = x^4/4 + C$, where C is an arbitrary constant.

Why C is mandatory: integration is the inverse of differentiation, and differentiation destroys constants — the derivative of any constant is 0. Hence $x^4/4$, $x^4/4 + 7$, $x^4/4 - 1/2$ and $x^4/4 + C$ for every real C all differentiate back to x^3 . The antiderivative of x^3 is therefore not a single function but an infinite family of functions differing only by a vertical shift. Writing only $x^4/4$ names one arbitrary member of that family and so answers a different question from the one asked; the "+ C " is what makes the answer the complete general solution, which is why an indefinite integral is called indefinite. Geometrically the family is a set of parallel curves, all having the same slope x^3 at each x , and C selects which one passes through a given point — exactly the role C plays when an initial condition is supplied.

Q4. Numerically evaluate $\int_2^6 8 dx$. Explain why it is a standard practice to omit the arbitrary constant (C) when solving the definite integral, even though it is required for indefinite integral. [5 marks · Jun 2026 Q2(d)]

Formula used — the fundamental theorem of calculus: $\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a)$, where $F'(x) = f(x)$

Here $f(x) = 8$ (a constant), $a = 2$, $b = 6$. An antiderivative is $F(x) = 8x$, since $d(8x)/dx = 8$.

$$\int_2^6 8 dx = [8x]_2^6 = 8(6) - 8(2) = 48 - 16 = 32$$

Cross-check geometrically: the region under the line $y = 8$ from $x = 2$ to $x = 6$ is a rectangle of height 8 and width $6 - 2 = 4$, so its area is $8 \times 4 = 32 \checkmark$

Answer: $\int_2^6 8 dx = 32$.

Why C is omitted in a definite integral: carry the constant through and watch it cancel. Taking the general antiderivative $F(x) + C$, $\int_2^6 8 dx = [8x + C]_2^6 = (8 \times 6 + C) - (8 \times 2 + C) = 48 + C - 16 - C = 48 - 16 = 32$. The constant appears once in the upper-limit term and once in the lower-limit term with opposite signs, so it cancels identically for every value of C . The value of a definite integral is therefore independent of which member of the antiderivative family is used, and writing C can never change the answer — so it is dropped as redundant. The contrast with the indefinite case is exactly this: an indefinite integral must describe the whole family of antiderivatives, and there C carries information; a definite integral is a single number obtained as a difference $F(b) - F(a)$, where that information is destroyed by the subtraction.

Q5. A sensor output is modelled by $f(x) = |x - 3|$. Check whether the function is continuous at $x = 3$. [5 marks · Jun 2026 Q5(c)]

Test to be applied, stated first. A function f is continuous at a point $x = a$ iff all three conditions hold: (i) $f(a)$ is defined, (ii) $\lim_{(x \rightarrow a)} f(x)$ exists (LHL = RHL, both finite), (iii) $\lim_{(x \rightarrow a)} f(x) = f(a)$.

Case form of the function: $|x - 3| = x - 3$ for $x \geq 3$, and $3 - x$ for $x < 3$.

Condition (i) — value at the point. $f(3) = |3 - 3| = |0| = 0$. Defined. ✓

Condition (ii) — the limit. Left-hand limit ($x < 3$, so $f(x) = 3 - x$): $\text{LHL} = \lim_{(x \rightarrow 3^-)} (3 - x) = 3 - 3 = 0$ Right-hand limit ($x > 3$, so $f(x) = x - 3$): $\text{RHL} = \lim_{(x \rightarrow 3^+)} (x - 3) = 3 - 3 = 0$ $\text{LHL} = \text{RHL} = 0$, both finite, so $\lim_{(x \rightarrow 3)} f(x) = 0$ exists. ✓

Condition (iii) — comparison. $\lim_{(x \rightarrow 3)} f(x) = 0$ and $f(3) = 0$, so they are equal. ✓

All three conditions are satisfied.

Answer: $f(x) = |x - 3|$ is continuous at $x = 3$.

The extra remark that earns the remaining marks: the function is continuous at $x = 3$ but not differentiable there. Checking the one-sided derivatives, left-hand derivative $= \lim_{(h \rightarrow 0^-)} [f(3 + h) - f(3)]/h = \lim_{(h \rightarrow 0^-)} |h|/h = -1$, right-hand derivative $= \lim_{(h \rightarrow 0^+)} |h|/h = +1$, and $-1 \neq +1$, so $f'(3)$ does not exist. Graphically the V-shaped graph has no break — the pen never leaves the paper, which is continuity — but it has a sharp corner at the vertex $(3, 0)$, where the slope jumps from -1 to $+1$, which is the failure of differentiability. Interpreted for the sensor: the output never jumps, so the reading is stable through $x = 3$, but its rate of change reverses abruptly at that point. Note also that continuity does not imply differentiability, though differentiability always implies continuity.

Q6. Verify whether $\lim_{(x \rightarrow 2)} |x - 2|/(x - 2)$ exists, supporting the answer with one-sided limits.

[practice]

Criterion: the limit exists iff $\text{LHL} = \text{RHL}$, both finite.

For $x > 2$: $|x - 2| = x - 2$, so $\text{RHL} = \lim_{(x \rightarrow 2^+)} (x - 2)/(x - 2) = \lim_{(x \rightarrow 2^+)} 1 = 1$

For $x < 2$: $|x - 2| = -(x - 2)$, so $\text{LHL} = \lim_{(x \rightarrow 2^-)} -(x - 2)/(x - 2) = \lim_{(x \rightarrow 2^-)} (-1) = -1$

$\text{LHL} = -1 \neq 1 = \text{RHL}$.

Answer: the limit does not exist.

Supporting argument: the function is the constant -1 to the left of 2 and the constant $+1$ to the right, so it approaches two different values depending on the direction of approach. Since a limit must be a single value approached from both sides, the two-sided limit fails to exist; the function has a jump discontinuity of size 2 at $x = 2$, and $f(2)$ is itself undefined ($0/0$).

Q7. Evaluate $\lim_{(x \rightarrow 3)} (x^2 - 9)/(x - 3)$. [practice]

Direct substitution gives $(9 - 9)/(3 - 3) = 0/0$, an indeterminate form, so the expression must be simplified first.

Factorise the numerator using $a^2 - b^2 = (a - b)(a + b)$: $x^2 - 9 = (x - 3)(x + 3)$

So, for $x \neq 3$, $(x^2 - 9)/(x - 3) = (x - 3)(x + 3)/(x - 3) = x + 3$

Now the limit: $\lim_{(x \rightarrow 3)} (x^2 - 9)/(x - 3) = \lim_{(x \rightarrow 3)} (x + 3) = 3 + 3 = 6$

Answer: 6.

Why the cancellation is valid: in evaluating $\lim_{(x \rightarrow 3)}$ the variable x approaches 3 but never equals 3, so $x - 3 \neq 0$ throughout and dividing by it is legitimate. The limit depends only on the values of the function near $x = 3$, not at $x = 3$ — which is why a limit can exist (here 6) even though the function is undefined at the point. Unlike the modulus example, both one-sided limits equal 6 here, so the limit exists; this is a removable discontinuity.

Q8. Using first principles, find the derivative of $f(x) = x^3$, and of $f(x) = 1/x$. [practice]

Definition: $f'(x) = \lim_{h \rightarrow 0} [f(x+h) - f(x)]/h$

Part 1 — $f(x) = x^3$. $f(x+h) = (x+h)^3 = x^3 + 3x^2h + 3xh^2 + h^3$
 $f(x+h) - f(x) = 3x^2h + 3xh^2 + h^3$
 $[f(x+h) - f(x)]/h = 3x^2 + 3xh + h^2$
 $f'(x) = \lim_{h \rightarrow 0} (3x^2 + 3xh + h^2) = 3x^2 + 0 + 0 = 3x^2$

Part 2 — $f(x) = 1/x$, $x \neq 0$. $f(x+h) = 1/(x+h)$
 $f(x+h) - f(x) = 1/(x+h) - 1/x = [x - (x+h)] / [x(x+h)] = -h / [x(x+h)]$
 $[f(x+h) - f(x)]/h = -1 / [x(x+h)]$
 $f'(x) = \lim_{h \rightarrow 0} -1/[x(x+h)] = -1/(x \cdot x) = -1/x^2$

Answer: $d(x^3)/dx = 3x^2$ and $d(1/x)/dx = -1/x^2$.

Why both results are consistent with the power rule: $d(x^n)/dx = n x^{n-1}$ gives $3x^2$ for $n = 3$, and for $1/x = x^{-1}$ it gives $(-1)x^{-2} = -1/x^2$, matching the first-principles work. In each part the single algebraic step that makes the limit computable is the cancellation of h — by factoring in Part 1 and by taking the LCM in Part 2 — because before it the quotient is of the indeterminate form $0/0$. Note the domain restriction $x \neq 0$ in Part 2, since $1/x$ is not defined at 0.

Q9. Evaluate $\int (x^4 + 3x) dx$ and explain the role of C . [practice]

Rules used: $\int [f(x) + g(x)]dx = \int f(x)dx + \int g(x)dx$ (sum rule), $\int k f(x)dx = k \int f(x)dx$ (constant multiple), and $\int x^n dx = x^{n+1}/(n+1) + C$ for $n \neq -1$.

$$\int (x^4 + 3x) dx = \int x^4 dx + 3 \int x dx = x^5/5 + 3 \cdot x^2/2 + C = x^5/5 + 3x^2/2 + C$$

$$\text{Verification: } d/dx [x^5/5 + 3x^2/2 + C] = (1/5)(5x^4) + (3/2)(2x) + 0 = x^4 + 3x \checkmark$$

Answer: $\int (x^4 + 3x) dx = x^5/5 + 3x^2/2 + C$.

Role of C : because the derivative of every constant is 0, infinitely many functions share the derivative $x^4 + 3x$, differing from one another only by an additive constant. The symbol C stands for that entire family, so the answer with C is the general antiderivative while the answer without it is one unnamed particular member. Only one constant is needed even though two terms were integrated, since the sum of two arbitrary constants is again a single arbitrary constant.

Q10. Evaluate $\int_1^5 6 dx$ and $\int_0^3 x^2 dx$, showing the $F(b) - F(a)$ step. [practice]

Rule used: $\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a)$, F being any antiderivative of f .

Part 1 — $\int_1^5 6 dx$. An antiderivative of 6 is $F(x) = 6x$. $\int_1^5 6 dx = [6x]_1^5 = 6(5) - 6(1) = 30 - 6 = 24$
Geometric check: a rectangle of height 6 and width $5 - 1 = 4$ has area 24 $\checkmark$

Part 2 — $\int_0^3 x^2 dx$. An antiderivative of x^2 is $F(x) = x^3/3$. $\int_0^3 x^2 dx = [x^3/3]_0^3 = 3^3/3 - 0^3/3 = 27/3 - 0 = 9$

Answer: $\int_1^5 6 dx = 24$ and $\int_0^3 x^2 dx = 9$.

Why no constant appears: had the antiderivative been written $F(x) + C$, the evaluation would read $(F(b) + C) - (F(a) + C)$, in which C cancels, so the numerical value is unaffected by the choice of antiderivative. Each result is also the area under the curve above the x -axis between the two limits — 24 for the horizontal line $y = 6$ over $[1, 5]$, and 9 for the parabola $y = x^2$ over $[0, 3]$ — which is the geometric meaning of a definite integral and a useful check on the sign and magnitude.

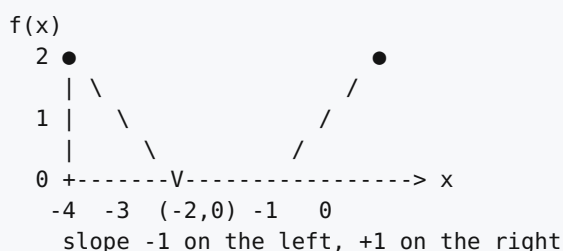
Q11. Check whether $f(x) = |x + 2|$ is continuous at $x = -2$, and whether it is differentiable there. [practice]

Case form: $|x + 2| = x + 2$ for $x \geq -2$, and $-(x + 2)$ for $x < -2$.

Continuity test at $x = -2$, three conditions: (i) $f(-2) = |-2 + 2| = |0| = 0$. Defined. ✓ (ii) $\text{LHL} = \lim_{(x \rightarrow -2^-)} -(x + 2) = -(-2 + 2) = 0$; $\text{RHL} = \lim_{(x \rightarrow -2^+)} (x + 2) = -2 + 2 = 0$. $\text{LHL} = \text{RHL} = 0$, so the limit exists and equals 0. ✓ (iii) $\lim_{(x \rightarrow -2)} f(x) = 0 = f(-2)$. ✓

All three hold, so f is continuous at $x = -2$.

Differentiability test at $x = -2$, using $f'(a) = \lim_{(h \rightarrow 0)} [f(a + h) - f(a)]/h$ with $f(-2) = 0$: Left-hand derivative = $\lim_{(h \rightarrow 0^-)} |h|/h = \lim_{(h \rightarrow 0^-)} (-h)/h = -1$ Right-hand derivative = $\lim_{(h \rightarrow 0^+)} |h|/h = \lim_{(h \rightarrow 0^+)} h/h = +1$ $-1 \neq +1$, so the limit defining $f'(-2)$ does not exist.



Answer: $f(x) = |x + 2|$ is continuous at $x = -2$ but not differentiable there.

Why both answers can hold at once: continuity only requires the graph to have no break at the point, which the common value 0 from both sides provides. Differentiability additionally requires a single well-defined tangent slope, and the V-shape has a corner at $(-2, 0)$ where the slope changes abruptly from -1 to $+1$, so no unique tangent exists. This illustrates that continuity does not imply differentiability, while differentiability does imply continuity.

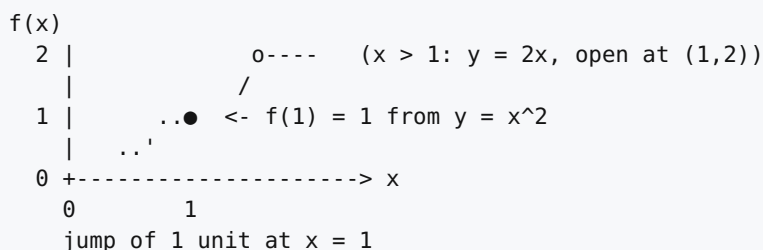
Q12. $f(x) = x^2$ for $x \leq 1$ and $f(x) = 2x$ for $x > 1$. Check continuity at $x = 1$. [practice]

Apply the three-condition continuity test at $x = 1$.

(i) Value at the point. Since $x = 1$ falls in the branch $x \leq 1$, $f(1) = 1^2 = 1$. Defined. ✓

(ii) The limit. $\text{LHL} = \lim_{(x \rightarrow 1^-)} x^2 = 1^2 = 1$ $\text{RHL} = \lim_{(x \rightarrow 1^+)} 2x = 2(1) = 2$ $\text{LHL} = 1$ and $\text{RHL} = 2$, so $\text{LHL} \neq \text{RHL}$ and $\lim_{(x \rightarrow 1)} f(x)$ does not exist. ✗

(iii) Cannot be tested, condition (ii) having already failed.



Answer: f is not continuous at $x = 1$ — it has a jump discontinuity there.

Which condition fails and why: the second condition of the continuity test fails, because the two one-sided limits are finite but unequal (1 from the left, 2 from the right), so no single limiting value exists at $x = 1$. The gap $\text{RHL} - \text{LHL} = 1$ is the jump. Note that condition (i) was satisfied — $f(1)$ is perfectly well defined as 1 — which shows that defining the function at the point is not sufficient

for continuity; all three conditions must hold. (The function would be continuous at $x = 1$ if the second branch were, say, $2x - 1$, since then $\text{RHL} = 1 = f(1)$.)